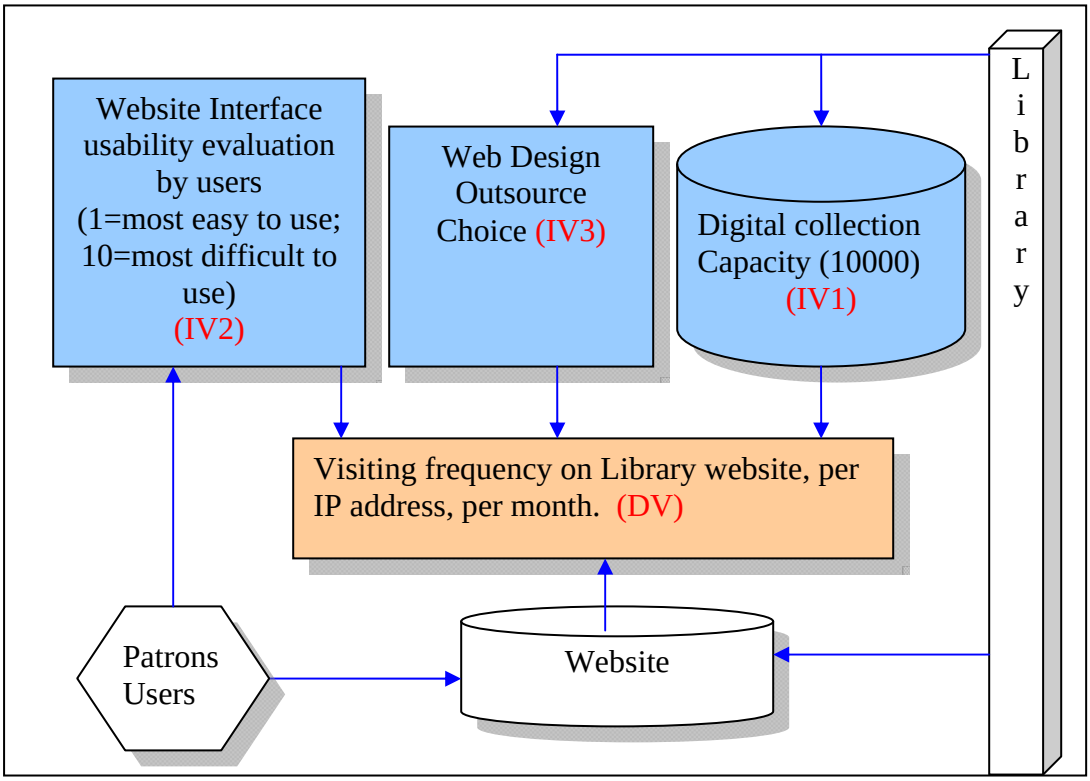
The Analysis of Variance on visiting frequency of library websites (Paper II)
By Andy Tang

1. Model Restatement

From the paper I, the conclusions are drawn that visiting frequency on Library website (DV) definitely are affected and influenced by 3 variables, which are separately “Digital Collection Capacity” (IV1), “Web Interface Usability Evaluation” (IV2), and “Web Design Outsource Choice” (IV3). However, how large is this kind of influence, and how big the difference between various means, remain the problem which should be solved.



On this paper II, a T-test of dependent variables (visiting frequency on Library website) against independent variables is made. The correlation between

different variables is calculated to measure the close degree of them. Furthermore, a multiple relationship is calculated to check whether the variability in DV can be explained by combination of IV1 and IV2.

2. T-Test of DV (visiting frequency on Library website) split by IV3 (Web Design Outsource Choice)

During the conversion from traditional library to digital library, we suppose that outsourcing web design task to some professional companies can bring a high visiting frequency on library website. In order to prove our conjecture, a null hypothesis is set up, which is H_0: μ1= μ2, equal to [H_0: μ1-μ2=0]. In this case, μ1 represents “visiting frequency on library website” under the condition that library outsource the web design task to professional companies. μ2 represents “visiting frequency on library website” under the situation that library chooses to design the website by themselves. At the same time, the alternative hypothesis are made, which is H_a: μ1> μ2 or [H_a: μ1-μ2>0].

The first step is to group the value of “visiting frequency on library website” based on which kind of decision the library makes on the aspect of outsourcing the web site design.

Visiting Frequency (Outsource Choice=Yes)	Visiting Frequency (Outsource Choice=No)
22	12
24	14
20	16
15	11
23	8
16	6
28	7
30	12
26	11

Visiting Frequency (Outsource Choice=Yes)	Visiting Frequency (Outsource Choice=No)
28	13
25	15
16	14
17	9
29	15
22	13

The second step is to calculate the two sample standard deviation.

S_1=5.05 and S_2=3.06

The third step is to calculate the pooled estimate of the common variance.

$$S^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2} = \frac{14(5.05)^2+14(3.06)^2}{15+15-2} = 17.42$$

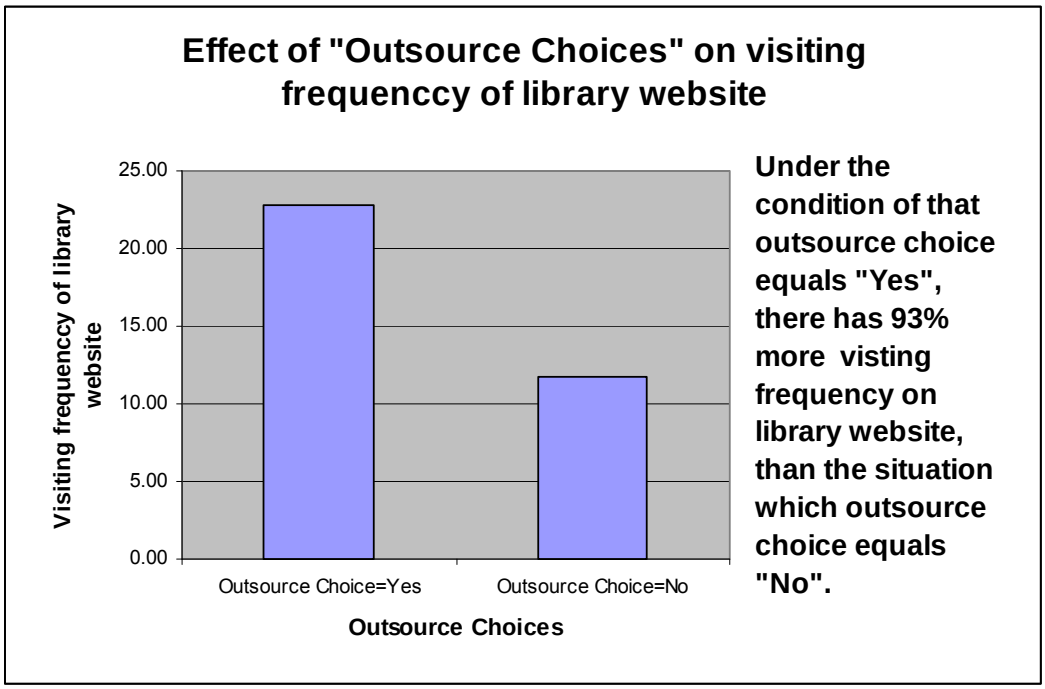
The fourth step is to calculate the student t statistic.

$$t = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{s^2(\frac{1}{n_1}+\frac{1}{n_2})}} = \frac{22.73-11.73}{\sqrt{17.42(\frac{1}{15}+\frac{1}{15})}} = 7.22$$

The alternative hypothesis: μ1> μ2 or, equivalently, [H_a: μ1-μ2>0] implies that we should use a one-tailed test in the upper tail of t distribution with 28 degrees of freedom. We can find the appropriate critical value for a rejection region with α=0.05 is 2.0484, and H_0 will be rejected if t>2.0484. For the reason of t=7.22, it is bigger compared with $t_{.05}$=2.0484. Hence, we reject the null hypothesis of H_0, which means there is a sufficient evidence to indicate that the outsource choice on the aspect of web design is superior at the 5% level of significance. At a word, whether choosing to outsource web design to

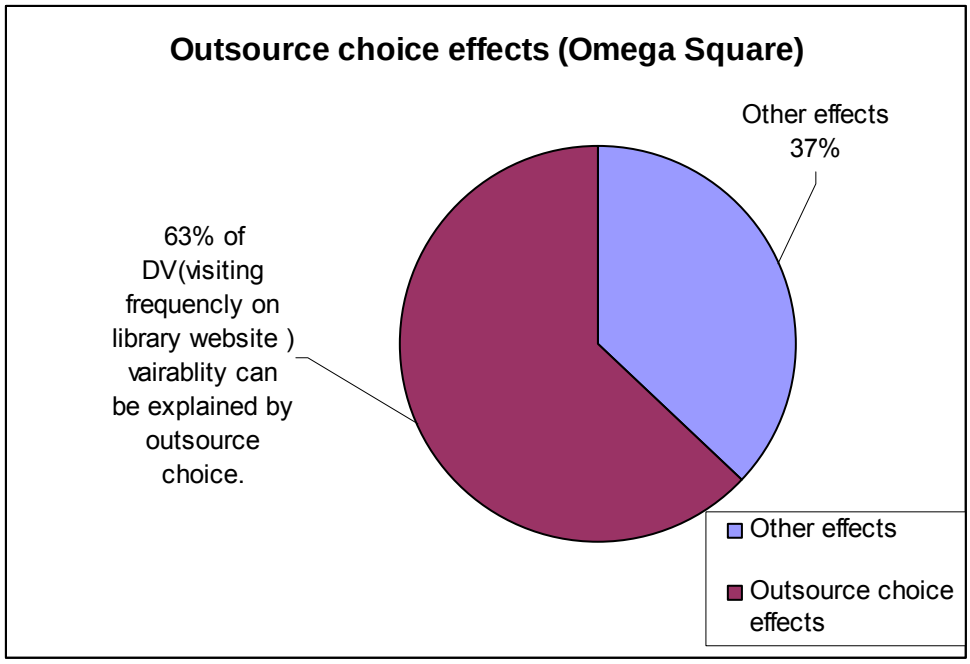
professional companies or not, definitely have a significant influence on user visiting frequency on library website.

Finally, the observed value of t for one-tailed test is t=7.22. Therefore, P-value= P (t>=2.0484) =0.0000000743. For the reason of that P-value is much smaller that .05, the results is reported to be significant.



Furthermore, an omega square is calculated, meaning that there are 63% variability on visiting frequency of library website can be explained by outsource choice.

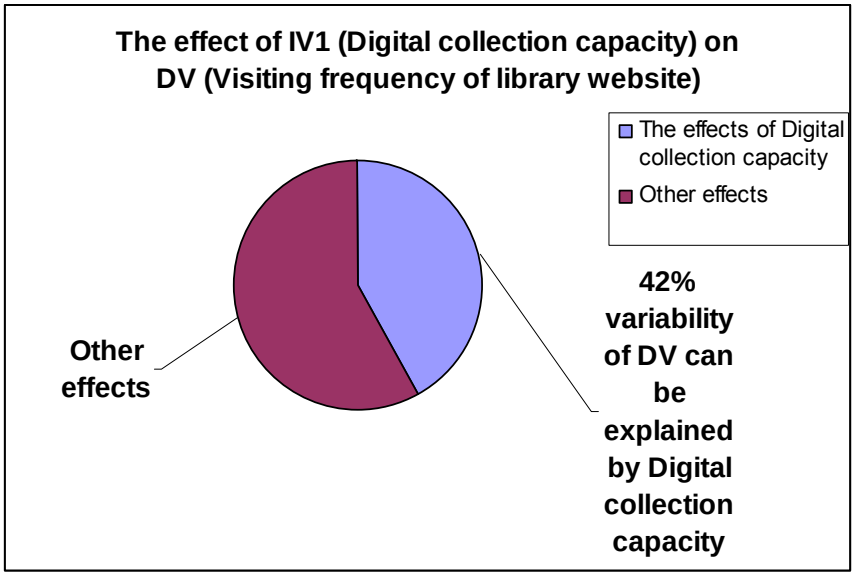
$$\omega^2 = \frac{t^2 - 1}{t^2 + n_1 + n_2 - 1} = \frac{7.22^2 - 1}{7.22^2 + 15 + 15 - 1} = 63\%$$



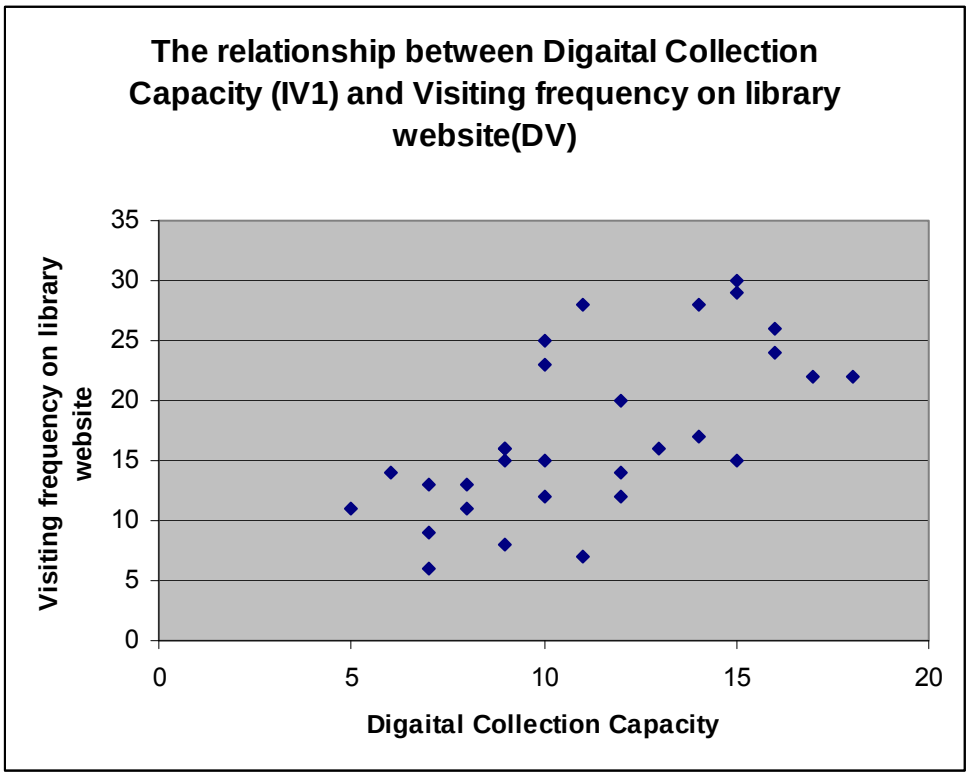
3. Two hypotheses on correlation

3.1. The relationship between DV and IV1

After calculation, the correlation r equals 0.65, indicating there is a significant correlation between DV (visiting frequency on Library website) and IV1 (Digital Collection Capacity). Therefore, r square is calculated equal 0.42.



For the reason of r becoming positive, the direction of relationship between DV and IV1 is alike. It means, when the digital collection capacity (IV1) increases, the visiting frequency on library website (DV) increases also. The trend is depicted in the illustration as below.

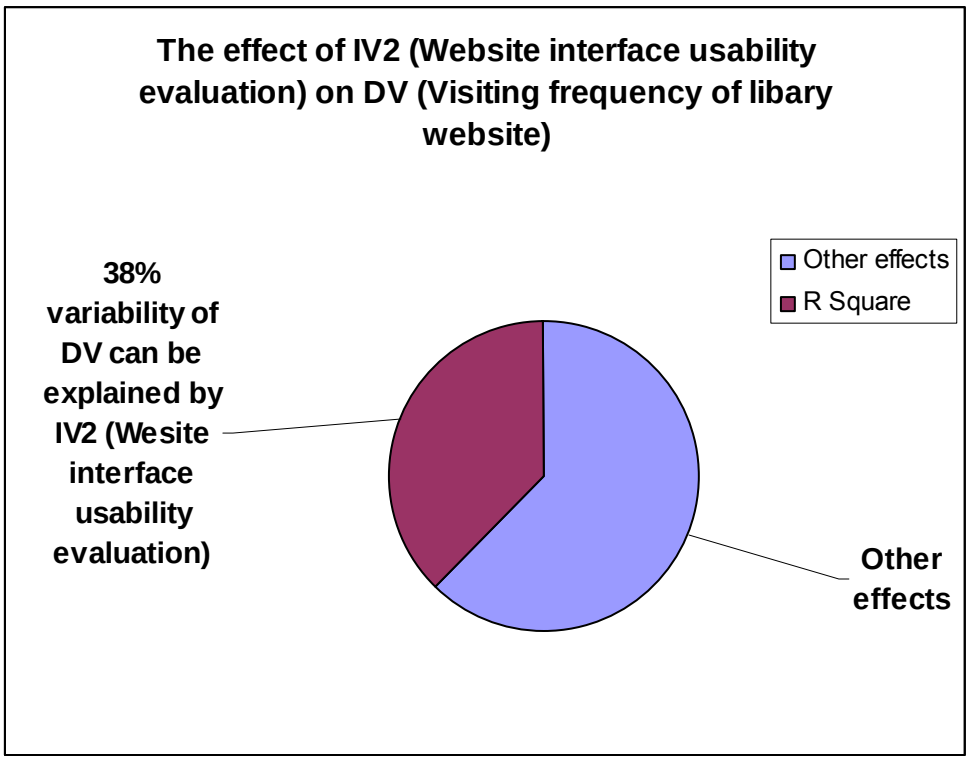


By using least-squares estimations, the formula for hypothesis between DV and IV1 are demonstrated as below. It displays that there is a linear positive correlation between variable of DV (visiting frequency on library website) and IV1 (digital collection capacity).

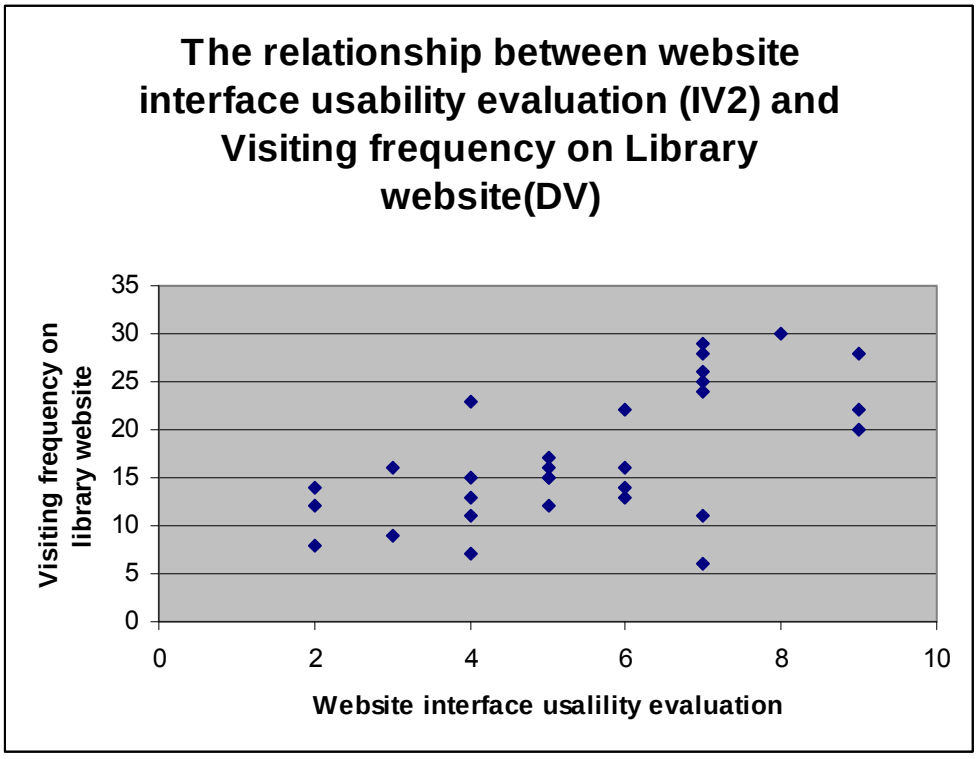
$$Y = 1.2814X + 2.9242$$

3.2. The relationship between DV and IV2

After calculation, the correlation r equals 0.62, clearly showing there is a significant correlation between DV (visiting frequency on Library website) and IV2 (website Interface usability evaluation by users). Moreover, r square is calculated, with the value equal 0.38.



Because r is positive, the direction of relationship between DV and IV2 is the same. It means, when the website interface usability evaluation (IV2) improves, the visiting frequency on library website (DV) increases also. The trend is displayed in the illustration as below.



By using least-squares estimations, the formula for hypothesis between DV and IV2 are demonstrated as below. Obviously, there is a linear positive correlation between variables of DV (visiting frequency on library website) and IV2 (website interface usability evaluation).

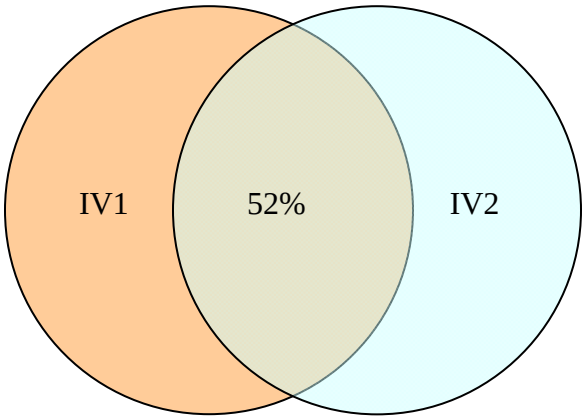
$$Y = 2.0864X + 5.7580$$

4. The relationship between IV1 (digital collection capacity) and IV2 (website interface usability evaluation)

The calculated result of r, which equals 0.52, seems to show there is a correlation between IV1 and IV2. The relevant r square equals 0.25. However, from the usual vision, this kind of correlation between IV1 and IV2 can not be explained in a reasonable way. From another vision, when users have a good impression on the large capacity of digital collections which library owns, they

may consider this library deals with electric archives very seriously, as a result, they may give a high score on the website interface usability. This may be the explanation why there is a correlation between IV1 (digital collection capacity) and IV2 (website interface usability evaluation). Sometimes, when the library website interface is very perfect and obtains a high score from users, it proves that the library puts big strength on electric archives making, usually leading a correction between IV1 and IV2.

The overlap between IV1 (digital collection capacity) and IV2 (website interface usability evaluation).



5. Multiple Relationship

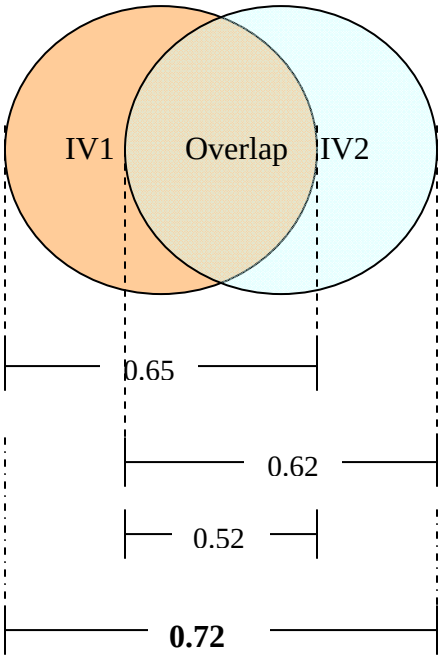
By using Excel, the multiple R is calculated and equal 0.72, and R square equal 0.52. After checking up the calculated results of r, we can find an overlap between IV1 (digital collection capacity) and IV2 (website interface usability evaluation), meaning that IV1 and IV2 are not independent to each other.

$$R_{IV1+IV2\rightarrow DV} = R_{IV1\rightarrow DV} + R_{IV2\rightarrow DV} - R_{IV1\leftrightarrow IV2}$$

$$0.72 \approx 0.65 + 0.62 - 0.52$$

The deduction can be demonstrated by the illustration as below.

The relationship between Multiple R and single r, proves that IV1 and IV2 are not independent of each other.



$$0.72 \approx 0.65 + 0.62 - 0.52$$

In theory, IVs should make unique, non-redundant contributions to explaining variability in DV. At the same time, IVs should be independent of each other. However, in reality, the absolute independence does not exist. The variables and everything in this world, more or less, have the overlap or interconnection among them. The only thing we can do in model building is to

decrease this kind of overlap as much as possible, by delicately selecting suitable concepts.

The formula of this multiple regression is demonstrated as below.

$$\boxed{Y = 0.8841X + 1.2969Y + 0.2279}$$

6. Conclusion

The t-test demonstrates that there is a strong correlation between IV3 (Web Design Outsource Choice) and DV (visiting frequency on Library website). The further calculation displays that whatever IV1 (Digital Collection Capacity), IV2 (Web Interface Usability Evaluation), they all have strong correlation with DV. However, the correlation between IV1 and IV2 shows that, IV1 and IV2 are not independent of each other. Moreover, the dependence of them definitely weakens their affection or influence on DV.

Sometimes, this kind of overlap between independent variables is inevitable. The best thing we can do is to analyze and clarify the scale of overlap between them. The ANOVA may be a better tool to study this kind of differences.